

A Falsification-Oriented Evaluation Framework for Deep Reinforcement Learning in Cross-Asset Portfolio Allocation

A TD3 case study under realistic trading constraints

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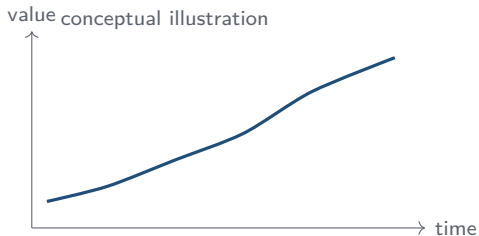
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Backtests can create false confidence

Apparent result

A DRL portfolio candidate can rank well after training and produce an attractive backtest.



Sources of overstatement

The credibility of the result depends on the evaluation design around the algorithm.

- Weak or mismatched benchmarks
- Simplified transaction costs
- Ambiguous cash treatment
- Candidate-search bias
- Limited statistical validation

Therefore, the thesis evaluates the claims around TD3.

The thesis separates three increasingly strong claims

Research question

Can a falsification-oriented evaluation framework distinguish ranking competitiveness, statistical credibility and practical feasibility in TD3-based cross-asset portfolio allocation?

1. Ranking

Performs well against matched benchmarks?

2. Statistics

Survives uncertainty and data-snooping controls?

3. Feasibility

Remains plausible under mandates and execution stress?

Interpretive rule

Ranking first does not prove statistical superiority, and statistical superiority would still not guarantee deployability.

The portfolio universe is compact and economically interpretable

SPY

equity growth

TLT

duration risk

GLD

real safe-haven
exposure

BTC-USD

digital alter-
native asset

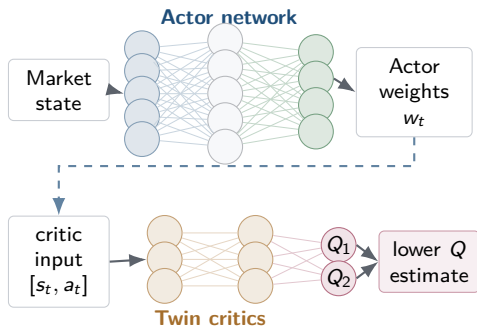
CASH / BIL

defensive liquidity and optionality

Core constraints

- Weekly allocation.
- Long-only; weights sum to one.
- Asset-specific transaction costs.
- Zero-CASH and BIL-CASH protocols.
- Walk-forward, multi-seed evaluation.

How TD3 produces portfolio allocations



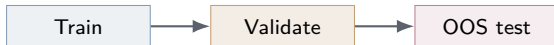
Why TD3 fits here

- Portfolio weights are continuous actions.
- Twin critics reduce overestimation risk.
- Target smoothing stabilizes critic targets.
- Actor updates are delayed for stability.

TD3 is evaluated under a demanding protocol

Evaluation Framework

- 1 Data and feature families.
- 2 TD3 candidate training.
- 3 Walk-forward validation and out-of-sample testing.
- 4 Matched deterministic benchmarks.
- 5 Bootstrap uncertainty and White Reality Check.
- 6 Regime, mandate, Pareto, execution and budget diagnostics.



Temporal separation is part of the test.

Protocol constants

Folds	4
Seeds	10
Budget	60 episodes
Benchmarks	14 rules
Cash	Zero-CASH / BIL-CASH

Same universe, cash protocol, cost schedule and evaluation window.

Ranking evidence remains competitive under matched assumptions

Zero-CASH

Metric	TD3	Trend SPY/CASH
Annualized return	8.69%	9.79%
Sharpe ratio	0.923	0.880
Maximum drawdown	-10.4%	-17.8%

Selected TD3: V3 clean macro cap 0.70.
Stronger risk profile, lower raw return.

BIL-CASH

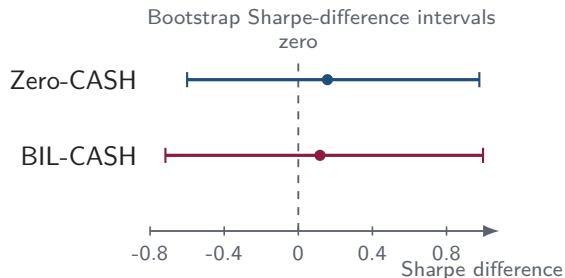
Metric	TD3	Trend SPY/CASH
Annualized return	10.65%	10.24%
Sharpe ratio	1.142	0.917
Maximum drawdown	-10.3%	-17.3%

Selected TD3: V7 macro+GARCH cap 0.80.
Stronger multi-metric ranking.

Boundary: cash treatment changes candidate selection and economic interpretation.

Selection note: Candidates were selected through the protocol’s diagnostic ranking; economic interpretation relies on standard metrics and statistical validation.

Statistical validation does not establish superiority

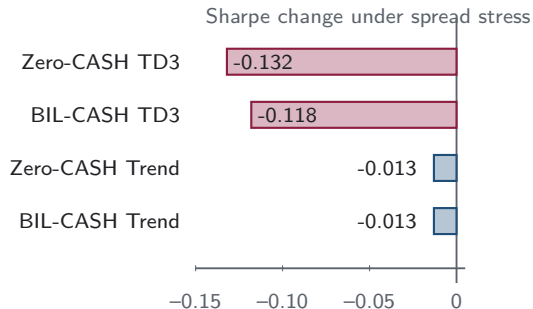


Protocol	Bootstrap CI	WRC p
Zero-CASH	$[-0.6011, 0.9767]$	0.714
BIL-CASH	$[-0.7172, 0.9963]$	0.677

Conclusion: TD3 remains ranking-competitive, but statistical superiority over the clean benchmark is not established.

Execution stress and mandates alter the investment reading

Execution-spread Sharpe degradation



Stress: additional half-spreads of 3/5/5/50/0 bps for SPY/TLT/GLD/BTC-USD/CASH, volatility-adjusted with $\beta = 0.5$; no retraining.

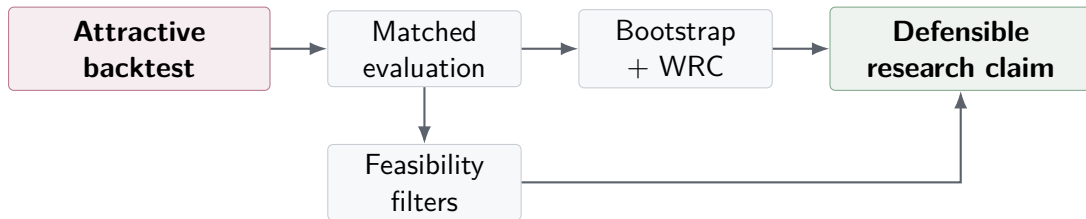
Mandate and practical filters

- Conservative hard filter: no strategy passes.
- Moderate/aggressive feasibility is conditional.
- Pareto competitiveness is tradeoff-dependent.
- Cash treatment changes candidate selection.

Only some claims survive

Claim	Reading	Evidence boundary
Ranking competitiveness	Supported	Selected TD3 candidates are competitive against the strongest matched benchmark on Sharpe and drawdown; return leadership depends on cash treatment.
Statistical superiority	Not supported	Bootstrap intervals include zero; WRC does not reject.
Cash assumption matters	Supported	Zero-CASH and BIL-CASH select different candidates.
Execution assumptions matter	Supported	TD3 histories degrade more under spread stress.
Conservative mandate feasibility	Not supported	No conservative candidate passes all hard filters.
Pareto competitiveness	Conditional	Near relevant tradeoff frontiers, profile-dependent.
Deterministic benchmarks remain credible	Supported	Clean Trend SPY/CASH remains a strong comparator.

The contribution



- Explicit cash treatment
- Transaction costs
- Matched deterministic benchmarks
- Bootstrap uncertainty
- Data-snooping control
- Practical feasibility filters

Contribution: disciplined framework for interpreting DRL portfolio evidence under realistic evaluation constraints.

Final answer: realistic evaluation narrows the claim

1. Ranking competitiveness **Supported**

Selected TD3 candidates remain competitive against matched deterministic benchmarks on risk-adjusted performance.

2. Statistical superiority **Not established**

Bootstrap Sharpe-difference intervals cross zero, and White Reality Check does not reject.

3. Practical feasibility **Conditional**

Cash treatment, execution stress, mandate constraints, Pareto trade-offs and market regimes materially affect interpretation.

TD3 is a competitive research candidate for dynamic allocation—not a statistically dominant trading strategy.

Yes. The framework distinguishes ranking competitiveness, statistical credibility and practical feasibility.

Main result: realistic evaluation materially changes what can responsibly be claimed from favourable DRL portfolio experiments.

Appendix: net returns include asset-specific costs

$$R_{t+1}^{\text{gross}} = \sum_i w_{i,t} r_{i,t+1}$$
$$C_t = \sum_i \kappa_i |w_{i,t} - \tilde{w}_{i,t-1}|$$
$$R_{t+1}^{\text{net}} = R_{t+1}^{\text{gross}} - C_t$$

Asset	Cost
SPY	2 bps
TLT	2 bps
GLD	2 bps
BTC-USD	10 bps
CASH Zero	0 bps
CASH BIL	2 bps

The same net-return environment is applied to TD3 and matched benchmarks; transaction costs are not a post-hoc penalty applied only to the learner.

Appendix: cash is not a footnote

Zero-CASH

Synthetic defensive sleeve with zero return and zero transaction cost. Useful for isolating allocation behaviour, but it ignores the opportunity cost of cash.

BIL-CASH

Investable short-Treasury ETF proxy with observable returns and ETF-like transaction cost. It changes the opportunity set and selected candidate.

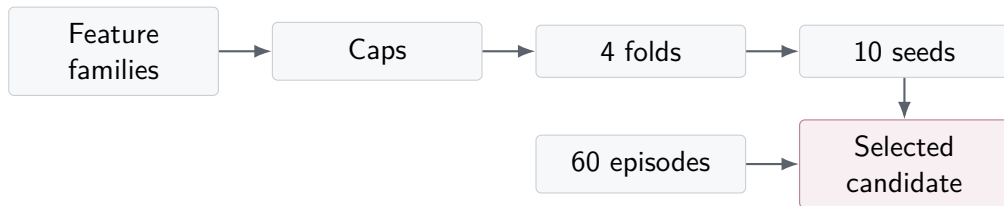
The two protocols answer different economic questions.

Appendix: TD3 Candidate Design and Feature Families

Family	Role	Interpretation note
V2 reference	Reference financial feature set.	Baseline rich state representation.
V3 clean macro	Audited macro variables aligned to the allocation frequency.	Used to test whether clean macro information improves allocation evidence.
V4 GARCH-style volatility	Volatility-state augmentation.	Tests whether fitted volatility information changes candidate behavior.
V5 no-volatility representation	Volatility-block ablation.	Tests whether simpler non-volatility states remain competitive.
V6 financial score-like indicators	Compact financial representation.	Variables are score-like indicators, not calibrated probabilities.
V7 macro + GARCH	Clean macro state combined with GARCH-style volatility.	Tests whether volatility augmentation changes the clean macro specification.
V8 EWMA/GARCH	EWMA/GARCH volatility hybrid.	Tests an alternative volatility-state construction.

Feature-family labels define candidate variants for evaluation, not standalone performance claims.

Appendix: candidate search is controlled by the protocol



The statistical layer then asks whether the selected result remains credible after uncertainty and candidate-search adjustment.

Appendix: Diagnostic selection score

Composite reporting tool: ranks evaluated candidates; it is not a standalone test of superiority.

$$S_{\text{diag}} = 0.30S_{\text{DSR}} + 0.20S_{\text{Sortino}} + 0.20S_{\text{Calmar}} \\ + 0.15S_{\text{DD}} + 0.10S_{\text{Stability}} \\ + 0.05S_{\text{Discipline}}.$$

Component	Wt.
DSR	30%
Sortino	20%
Calmar	20%
Drawdown	15%
Stability	10%
Discipline	5%

Implementation details

- Components are bounded to $[0, 1]$; Sortino, Calmar and drawdown are comparison-set normalized.
- DSR assumes 25 effective trials, with 10 and 50 used as sensitivity checks.
- Stability: 70% inverse Sharpe dispersion and 30% worst-drawdown performance.
- Discipline: 50% cash, 30% turnover and 20% diversification.
- Computed ex post; it does not alter TD3 training or returns.

Selection $\neq$ inference: bootstrap and WRC test inference; the score only orders candidates.

Appendix: benchmark families are economically interpretable

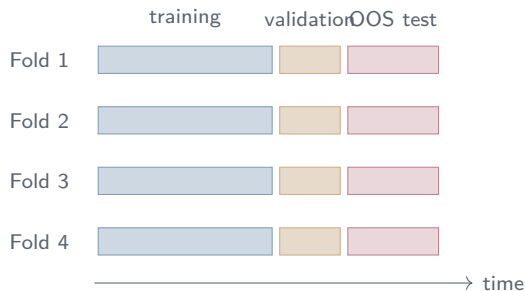
Passive and static comparators

- Buy-and-hold sleeves
- 60/40 SPY/TLT
- Equal weight
- Equal weight risky

Dynamic rules

- Defensive risk-off
- Momentum winner
- Risk-adjusted momentum
- Trend SPY/CASH
- Rolling Markowitz, min-var and risk parity

Appendix: walk-forward evaluation enforces temporal discipline



Folds

4

Seeds

10

Temporal ordering prevents the policy from being evaluated on periods it used for training decisions.

Appendix: Bootstrap and WRC deepen the statistical test

1. Pairwise bootstrap

Tests whether TD3–Trend Sharpe differences are distinguishable from zero within each cash protocol.

2. White Reality Check

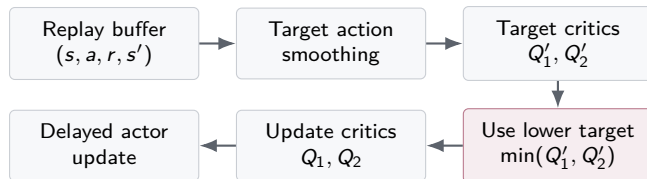
Applies the stricter candidate-set lens after feature families, caps, folds and seeds have been searched.

The statistical layer converts a ranking lead into a weaker conclusion: competitive, but not statistically dominant.

Protocol	Bootstrap CI	WRC p	Result
Zero-CASH	crosses 0	0.714	fail to reject
BIL-CASH	crosses 0	0.677	fail to reject

- Bootstrap: uncertainty remains too wide for a dominance claim.
- WRC: high p -values preserve the non-rejection after candidate search.

Appendix: TD3 internal update loop

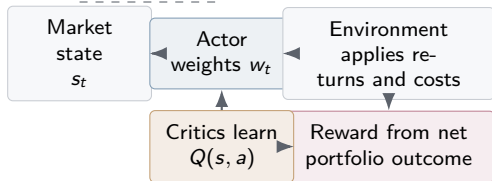


- Actor proposes continuous portfolio weights.
- Two critics reduce overestimation risk.
- Target smoothing stabilizes critic targets.
- Actor updates are delayed relative to critic updates.

No network architecture or hyperparameter values are claimed on this slide.

Appendix: reward design turns outcomes into learning signals

no direct return forecast target



$$r_{p,t+1}^{net} = w_t' r_{t+1} - \text{costs}_t$$
$$R_{t+1} = f(r_{p,t+1}^{net}, \text{risk terms})$$

What this defends

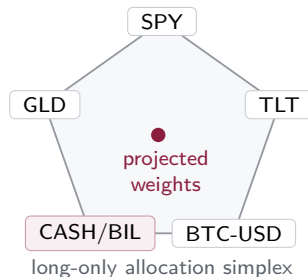
- TD3 optimizes a policy, not a price forecast.
- Costs enter before the learning signal.
- Drawdown penalties are modelling assumptions.

Appendix: feasible-action projection keeps weights investable

$$w_{i,t} \geq 0$$
$$\sum_i w_{i,t} = 1$$

The policy is evaluated inside a long-only, fully invested portfolio environment, including the defensive sleeve.

Conceptual representation of the feasible allocation set



Appendix: regime analysis checks where TD3 wins

Protocol	TD3 appearances	Recurring benchmark winners
Zero-CASH	V8 TD3 uncapped appears in AI recovery and 2023 cells; V7 TD3 cap 0.80 appears in 2024 cells.	BuyHold BTC/GLD/SPY, rolling Markowitz and rolling min-var remain strong.
BIL-CASH	There is no TD3 model regime-grid winner.	Risk-adjusted momentum dominates most Sharpe, max-DD, mandate and composite cells.
Stress slices	TD3 does not dominate COVID, inflation or recent-window summaries.	BTC wins risk-on shock; GLD wins inflation; momentum wins late windows.

Interpretation

- Regime results reject a single universal winner.
- Cash treatment changes which strategy families survive.
- TD3 is useful in selected states, not across all states.

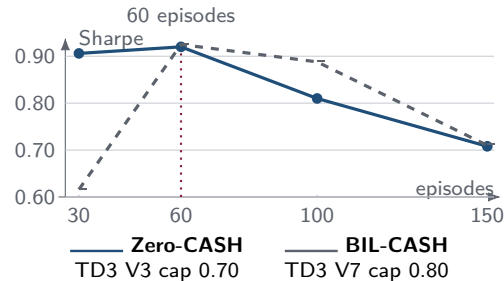
Regime evidence supports conditional competitiveness, not regime-invariant dominance.

Appendix: mandate filters distinguish ranking from suitability

Conservative	No strategy passes all hard filters
Moderate	Conditional feasibility
Aggressive	Conditional feasibility
Pareto reading	Competitive, tradeoff-dependent

Standard rankings do not determine investor suitability. A candidate can rank well and still fail a mandate.

Appendix: training-budget convergence is not obviously undertrained



60 episodes is not flagged as obviously undertrained.

Diagnostic reading

- 60 episodes sits near the observed Sharpe peak for both selected pairs.
- Longer budgets do not mechanically improve the selected candidates.
- This is a convergence diagnostic, not a proof of global optimality.

The fixed budget is defensible as a pre-set protocol choice, not an ex-post tuning lever.

Appendix: scope limits define the boundary of the conclusions

-
- Compact U.S.-centric asset universe.
 - Historical sample is finite and contains regime-specific episodes.
 - Costs are transparent modelling assumptions, not account-specific execution estimates.
 - No taxes, investor liabilities, custody constraints or live trading deployment.
 - Statistical validation limits claims but does not guarantee future performance.

Appendix: reproducibility makes the evaluation auditable

Evidence the panel can audit

- Asset universe and cash-protocol definitions
- Walk-forward split and fold construction
- Candidate grid, training budget and seed list
- Matched benchmark implementations
- Transaction-cost and execution-stress assumptions

The contribution is not a single favourable TD3 backtest; it is an evaluation design that a reader can reconstruct, inspect and challenge.

Project repository

`github.com/luiguiHerrera/
portfolio_drl_td3`